BIN CHEN

Address: 2233 Tech Drive, Mudd Room 4706, Evanston, IL 60208

Phone: 480-310-8682 ⋄ Email: bin.chen@northwestern.edu ⋄ Website: binchen.ac

EDUCATION

Arizona State University, Tempe, AZ — 2014 – 2018
Ph.D. Materials Science and Engineering

Fuzhou University, China — 2010 – 2014
B.Eng. Materials Science and Engineering

RESEARCH EXPERIENCE

Northwestern University, Evanston, IL — 2025 – present
Director of Research, Paula M. Trienens Institute for Sustainability and Energy

Northwestern University, Evanston, IL — 2024 – present
Associate Director, TEAMUP Consortium (Tandems for Efficient and Advanced Modules using Ultra-stable Perovskites)

Northwestern University, Evanston, IL — 2022 – present
Research Associate Professor (2024 – present); Research Assistant Professor (2022 – 2024), Department of Chemistry; Department of Electrical & Computer Engineering
Sponsoring Principal Investigator: Prof. Edward H. Sargent
Research: Emerging semiconductors for electronics and optoelectronics

University of Toronto, Canada — 2018 – 2022
Postdoctoral Fellow, Department of Electrical & Computer Engineering
Advisor: Prof. Edward H. Sargent
Research: Perovskite-based tandem solar cells and quantum dot infrared photodetectors

Arizona State University, Tempe, AZ — 2014 – 2018
Graduate Research Associate, School for Engineering of Matter, Transport & Energy
Advisor: Prof. Sefaattin Tongay
Dissertation: Atomic Scale Characterizations of Two-dimensional Anisotropic Materials and Their Heterostructures
Committee Members: Prof. Sefaattin Tongay, Prof. Mariana Bertoni, Prof. Shery Chang

SELECTED PUBLICATIONS

† equal contribution, * corresponding

25. Shin, D.†; Kitade, S.†; Duan, L.†; Liu, C.†; Li, C.†; Huang, C.; Kazianga, U. H.; López-Arteaga, R.; Reitz, S. A.; Choi, D.; Yang, Y.; Kim, H. J.; Lee, J.; Beights, J.; Kwon, H. W.; Song, J.-Y.; Liu, Y.; Choi, J.; Mirkin, C. A.; Kanatzidis, M. G.; **Chen, B.***; Sargent, E. H.* Sterically Gated Lewis Acid and Base Pairs Enable Orthogonal Defect Passivation in Perovskite Solar Cells. *Nat. Chem.* **2026**, in press. https://doi.org/10.1038/s41557-026-02261-z.

24. Shi, P.†; Zhuang, J.†; Zhang, D.†; Li, C.†; Zeiske, S.; Hou, J.; Gilley, I. W.; Yavuz, I.; Kazianga, U.; Li, C.; Zhang, Y.; Huang, K.; Itsoponpan, T.; Jiang, X.; Chen, C.-H.; Huang, C.; Yang, Y.; Zhang, X.; Nandi, P.; Reitz, S. A.; Facchetti, A.; White, K. P.; Wyatt, K.; Toney, M. F.; Kanatzidis, M. G.; Marks, T. J.*; Liu, C.*; **Chen, B.***; Sargent, E. H.* Phase-Homogeneous Mixed Halide Perovskites for Stable Tandem Photovoltaics. *Nature* **2026**. https://doi.org/10.1038/s41586-026-10929-2.

23. Hou, J.; Gilley, I. W.; Wiggins, T. E.; Liu, C.; Yang, Y.; **Chen, B.***; Kanatzidis, M. G.*; Sargent, E. H.* Synthetic Routes to Advancing Perovskite Solar Cells Through Interface Design. *Nat. Synth.* **2026**, *5*, 975–988.

22. Zeiske, S.; Ban, H. W.; Li, X.; Deng, B.; López-Arteaga, R.; Kazianga, U. H.; Han, M. G.; Kim, T.-G.; **Chen, B.***; Sargent, E. H.* Revealing Additional Size-Dependent Defect Suppression Channels Governing Detectivity in InAs Colloidal Quantum Dot Photodiodes. *Nano Lett.* **2025**, *25* (48), 16862–16868.

21. Huang, C.; Yang, Y.; Liu, C.; Chen, H.; Chaudhuri, S.; Jeon, W. C.; Li, M.; Rolston, N.; Bati, A. S. R.; Gilley, I. W.; Kumral, B.; Serles, P.; Filleter, T.; Schatz, G. C.; Kanatzidis, M. G.; **Chen, B.***; Chen, L. X.*; Sargent, E. H.* Electrostatically Enhanced Buried Interface Binding of Self-Assembled Monolayers for Efficient and Stable Inverted Perovskite Solar Cells. *Adv. Mater.* **2025**, *37* (43), e08740.

20. Bati, A. S. R.; Liu, C.; Gilley, I. W.; Musgrave, C. B., III; Maxwell, A.; Steele, J. A.; Yang, Y.; Chen, H.; Wan, H.; Xu, J.; Solano, E.; Zhang, R.; Huang, C.; Rehl, B.; Lempesis, N.; Carnevali, V.; Vezzosi, A.; Zeng, L.; Grater, L.; Li, M.; Rolston, N.; Choi, D.; Sláma, V.; Rothlisberger, U.; Wang, L.; Goddard, W. A., III; Kanatzidis, M. G.; **Chen, B.***; Bakr, O. M.*; Sargent, E. H.* A Chemically Bonded Monolayer Interface Enables Enhanced Thermal Stability and Efficiency in Pb-Sn Perovskite Solar Cells. *Joule* **2025**, *9* (9), 102047.

19. Choi, D.; Shin, D.; Li, C.; Liu, Y.; Bati, A. S. R.; Kachman, D. E.; Yang, Y.; Li, J.; Lee, Y. J.; Li, M.; Penukula, S.; Kim, D. B.; Shin, H.; Chen, C.-H.; Park, S. M.; Liu, C.; Maxwell, A.; Wan, H.; Rolston, N.; Sargent, E. H.*; **Chen, B.*** Carboxyl-Functionalized Perovskite Enables ALD Growth of a Compact and Uniform Ion Migration Barrier. *Joule* **2025**, 9 (3), 101801.

18. Yang, Y.; Chen, H.; Liu, C.; Xu, J.; Huang, C.; Malliakas, C. D.; Wan, H.; Bati, A. S. R.; Wang, Z.; Reynolds, R. P.; Gilley, I. W.; Kitade, S.; Wiggins, T. E.; Zeiske, S.; Suragtkhuu, S.; Batmunkh, M.; Chen, L. X.; **Chen, B.***; Kanatzidis, M. G.*; Sargent, E. H.* Amidination of Ligands for Chemical and Field-Effect Passivation Stabilizes Perovskite Solar Cells. *Science* 2024, *386* (6724), 898–902.

17. Li, C.†; Chen, L.†; Jiang, F.†; Song, Z.†; Wang, X.; Balvanz, A.; Ugur, E.; Liu, Y.; Liu, C.; Maxwell, A.; Chen, H.; Liu, Y.; Wang, Z.; Xia, P.; Li, Y.; Fu, S.; Sun, N.; Grice, C. R.; Wu, X.; Fink, Z.; Hu, Q.; Zeng, L.; Jung, E.; Wang, J.; Park, S. M.; Luo, D.; Chen, C.; Shen, J.; Han, Y.; Perini, C. A. R.; Correa-Baena, J.-P.; Lu, Z.-H.; Russell, T. P.; De Wolf, S.; Kanatzidis, M. G.; Ginger, D. S.; **Chen, B.***; Yan, Y.*; Sargent, E. H.* Diamine Chelates for Increased Stability in Mixed Sn–Pb and All-Perovskite Tandem Solar Cells. *Nat. Energy* 2024, *9* (11), 1388–1396.

16. Chen, H.†; Liu, C.†; Xu, J.†; Maxwell, A.†; Zhou, W.†; Yang, Y.; Zhou, Q.; Bati, A. S. R.; Wan, H.; Wang, Z.; Zeng, L.; Wang, J.; Serles, P.; Liu, Y.; Teale, S.; Liu, Y.; Saidaminov, M.; Hoogland, S.; Filleter, T.; Kanatzidis, M. G.; **Chen, B.***; Ning Z*; Sargent, E. H.* Improved charge extraction in inverted perovskite solar cells with dual-site-binding ligands. *Science* 2024, 384 (6692), 189–193.

15. Xu, J.; Maxwell, A.; Song, Z.; Bati, A. S. R.; Chen, H.; Li, C.; Park, S. M.; Yan, Y.; **Chen, B.***; Sargent, E. H.* The Dynamic Adsorption Affinity of Ligands Is a Surrogate for the Passivation of Surface Defects. *Nat. Commun.* 2024, 15 (1), 2035.

14. Maxwell, A.†; Chen, H.†; Grater, L.; Li, C.; Teale, S.; Wang, J.; Zeng, L.; Wang, Z.; Park, S. M.; Vafaie, M.; Sidhik, S.; Metcalf, I. W.; Liu, Y.; Mohite, A. D.; **Chen, B.***; Sargent, E. H.* All-Perovskite Tandems Enabled by Surface Anchoring of Long-Chain Amphiphilic Ligands. *ACS Energy Lett.* 2024, *9*, 520–527.

13. Yang, Y.†; Liu, C.†; Ding, Y.†; Ding, B.†; Xu, J.†; Liu, A.; Yu, J.; Grater, L.; Zhu, H.; Hadke, S. S.; Sangwan, V. K.; Bati, A. S. R.; Hu, X.; Li, J.; Park, S. M.; Hersam, M. C.; **Chen, B.***;

Nazeeruddin, M. K.*; Kanatzidis, M. G.*; Sargent, E. H.* A Thermotropic Liquid Crystal Enables Efficient and Stable Perovskite Solar Modules. *Nat. Energy* 2024, *9*, 316–323.

12. Liu, C.†; Yang, Y.†; Chen, H.†; Xu, J.†; Liu, A.†; Bati, A. S. R.; Zhu, H.; Grater, L.; Hadke, S. S.; Huang, C.; Sangwan, V. K.; Cai, T.; Shin, D.; Chen, L. X.; Hersam, M. C.; Mirkin, C. A.; **Chen, B.***; Kanatzidis, M. G.*; Sargent, E. H.* Bimolecularly-passivated interface enables efficient and stable inverted perovskite solar cells, *Science* 2023, 382 (6672), 810–815.

11. Liu, Y.†; Zhu, T.†; Grater, L.†; Chen, H.†; Reis, R.; Maxwell, A.; Cheng, M.; Dong, Y.; Teale, S.; Leontowich, A. F. G.; Kim, C.; Chan, P. T.; Wang, M.; Paritmongkol, W.; Gao, Y.; Park, S.; Xu, J.; Khan, J. I.; Laquai, F.; Walker, G. C.; Dravid, V. P.; **Chen, B.***; Sargent, E. H.* A Three-Dimensional Quantum Dot Network Stabilizes Perovskite Solids via Hydrostatic Strain. *Matter* 2024, 7 (1), 107–122.

10. Chen, H.†; Maxwell, A.†; Li, C.†; Teale, S.†; **Chen, B.†**; Zhu, T.; Ugur, E.; Harrison, G.; Grater, L.; Wang, J.; Wang, Z.; Zeng, L.; Park, S. M.; Chen, L.; Serles, P.; Awni, R. A.; Subedi, B.; Zheng, X.; Xiao, C.; Podraza, N. J.; Filleter, T.; Liu, C.; Yang, Y.; Luther, J. M.; De Wolf, S.; Kanatzidis, M. G.; Yan, Y.; Sargent, E. H. Regulating Surface Potential Maximizes Voltage in All-Perovskite Tandems. *Nature* 2023, 613 (7945), 676–681.

9. Chen, H.†; Teale, S.†; **Chen, B.†**; Hou, Y.†; Grater, L.; Zhu, T.; Bertens, K.; Park, S. M.; Atapattu, H. R.; Gao, Y.; Wei, M.; Johnston, A. K.; Zhou, Q.; Xu, K.; Yu, D.; Han, C.; Cui, T.; Jung, E. H.; Zhou, C.; Zhou, W.; Proppe, A. H.; Hoogland, S.; Laquai, F.; Filleter, T.; Graham, K. R.; Ning, Z.; Sargent, E. H. Quantum-Size-Tuned Heterostructures Enable Efficient and Stable Inverted Perovskite Solar Cells. *Nat. Photonics* 2022, 16 (5), 352–358.

8. **Chen, B.**; Sargent, E. H. What Does Net Zero by 2050 Mean to the Solar Energy Materials Researcher? *Matter* 2022, 5 (5), 1322–1325.

7. **Chen, B.**; Chen, H.; Hou, Y.; Xu, J.; Teale, S.; Bertens, K.; Chen, H.; Proppe, A.; Zhou, Q.; Yu, D.; Xu, K.; Vafaie, M.; Liu, Y.; Dong, Y.; Jung, E. H.; Zheng, C.; Zhu, T.; Ning, Z.; Sargent, E. H. Passivation of the Buried Interface via Preferential Crystallization of 2D Perovskite on Metal Oxide Transport Layers. *Adv. Mater.* 2021, e2103394.

6. Fang, Z.†; Wang, L.†; Mu, X.†; **Chen, B.†**; Xiong, Q.; Wang, W. D.; Ding, J.; Gao, P.; Wu, Y.; Cao, J. Grain Boundary Engineering with Self-Assembled Porphyrin Supramolecules for Highly Efficient Large-Area Perovskite Photovoltaics. *J. Am. Chem. Soc.* 2021.

5. Jung, E. H.†; **Chen, B.†**; Bertens, K.; Vafaie, M.; Teale, S.; Proppe, A.; Hou, Y.; Zhu, T.; Zheng, C.; Sargent, E. H. Bifunctional Surface Engineering on SnO2 Reduces Energy Loss in Perovskite Solar Cells. *ACS Energy Lett.* 2020, 5 (9), 2796–2801.

4. **Chen, B.**; Baek, S.-W.; Hou, Y.; Aydin, E.; De Bastiani, M.; Scheffel, B.; Proppe, A.; Huang, Z.; Wei, M.; Wang, Y.-K.; Jung, E.-H.; Allen, T. G.; Van Kerschaver, E.; García de Arquer, F. P.; Saidaminov, M. I.; Hoogland, S.; De Wolf, S.; Sargent, E. H. Enhanced Optical Path and Electron Diffusion Length Enable High-Efficiency Perovskite Tandems. *Nat. Commun.* 2020, 11 (1), 1257.

3. Manekkathodi, A.†; **Chen, B.†**; Kim, J.; Baek, S.-W.; Scheffel, B.; Hou, Y.; Ouellette, O.; Saidaminov, M. I.; Voznyy, O.; Madhavan, V. E. Solution-Processed Perovskite-Colloidal Quantum Dot Tandem Solar Cells for Photon Collection beyond 1000 Nm. *Journal of Materials Chemistry A* 2019, 7 (45), 26020–26028.

2. **Chen, B.**; Wu, K.; Suslu, A.; Yang, S.; Cai, H.; Yano, A.; Soignard, E.; Aoki, T.; March, K.; Shen, Y. Controlling Structural Anisotropy of Anisotropic 2D Layers in Pseudo-1D/2D Material Heterojunctions. *Adv. Mater.* 2017, 29 (34), 1701201.

1. **Chen, B.**; Sahin, H.; Suslu, A.; Ding, L.; Bertoni, M. I.; Peeters, F.; Tongay, S. Environmental Changes in MoTe2 Excitonic Dynamics by Defects-Activated Molecular Interaction. *ACS nano*

2015, 9 (5), 5326–5332.

FUNDING SUPPORT

Pending Research

3. HPC4EI Spring 2026
   U.S. Department of Energy, High Performance Computing for Energy Innovation Program
   $200,000 (Northwestern share)
   Full proposal submitted 08/2026; award notification expected 10/2026
   Physics-Aware Discovery of Stable Perovskite–Molecular Interfaces for Durable U.S.-Manufactured Tandem Photovoltaic Modules
   Role: Principal Investigator

2. NSF 23-612
   U.S. National Science Foundation
   $600,000
   Resubmitted 01/2026
   Synthesis and surface chemistry of InAs colloidal quantum dots toward efficient short-wave infrared detection
   Role: Co-Principal Investigator

1. DE-FOA-0003337, Photovoltaics Research and Development (PVRD) 2024
   U.S. Department of Energy, Office of Energy Efficiency and Renewable Energy
   $1,500,000
   Full application submitted 09/2024
   Efficient and stable near-infrared organic photovoltaics for hybrid organic tandem solar cells
   Role: Co-Principal Investigator

Current Research

3. DOI-BAA-SOLSTICE-FY25-01
   Intelligence Advanced Research Projects Activity (IARPA)
   $600,000 (Northwestern share)
   06/2026 – 12/2027
   Superior Options for Long-life Solar Technologies with Impressive Conversion Efficiencies (SOLSTICE)
   Role: Co-Principal Investigator

2. DE-EE0010502
   U.S. Department of Energy, Office of Energy Efficiency and Renewable Energy
   $9,000,000
   09/2023 – 10/2025; relaunched 07/2026 – 09/2027
   TEAMUP: Tandems for Efficient and Advanced Modules using Ultrastable Perovskites
   Role: Senior Personnel; Associate Director (2024 – present)

1. CRG2024
   King Abdullah University of Science and Technology
   $180,000 (Northwestern share)
   2025 – 2028
   PRENNIAL
   Role: Co-Principal Investigator

Completed Research

7. DE-FOA-0003058
   U.S. Department of Energy, Office of Energy Efficiency and Renewable Energy

$500,000 (Northwestern share)
Awarded 09/2024 – 08/2026; terminated by the sponsor 10/2025 in the government-wide DOE project terminations
STACKED: Stability and Characterization of Hole-Transporting Layers Key to Enabling Outdoor Durability
Role: Co-Principal Investigator

6. HR001122S0044-SNAP-FP-009
U.S. Department of Defense, Defense Advanced Research Projects Agency
$10,000,000
Awarded 08/2023 – 07/2027; closed out 11/2025
SYNCED: Interfacing Synthetic Biology with Electrochemical Detectors for Smart Non-Invasive Assays of Physiology
Role: Senior Personnel

5. OSR-5624
King Abdullah University of Science and Technology
$350,000
02/2024 – 01/2026
All-Perovskite Tandem Solar Cells
Role: Co-Principal Investigator

4. Seed Funding Initiative
Center for Engineering Sustainability and Resilience (CESR), Northwestern University
$80,000
02/2024 – 08/2025
Introducing AC Photo-Hall Method: Separating Electron/Hole Mobilities in Perovskite Photovoltaics
Role: Co-Principal Investigator

3. Translation and Incubation Fund
Paula M. Trienens Institute for Sustainability and Energy, Northwestern University
$100,000
01/2024 – 12/2024
Stable perovskite solar cells with cost-effective bilayer metal oxides as electron transport layers
Role: Principal Investigator

2. N00014-20-1-2572
U.S. Department of the Navy, Office of Naval Research
$480,000
08/2020 – 07/2023
Wide-bandgap perovskites for efficient, stable tandems
Role: Senior Personnel

1. OSR-2020-CRG9-4350.2
King Abdullah University of Science and Technology, Office of Sponsored Research
$600,000
04/2021 – 03/2024
SOLSTICE: Solar-driven Circular Carbon Enabled by Perovskite/Perovskite/Si Triple-Junction Tandems
Role: Senior Personnel

PATENTS

3. US Patent Application: Passivating Perovskite Optoelectronic Devices (Application No. 19/375,634, 2026)

2. US and Canadian Patent: A Surface Treatment Method to Passivate Inverted Structure Perovskite Solar Cells
1. US Patent Application: Perovskite Solar Cells With Dual Site Binding Ligands

INVITED TALKS

22. MRS Fall Meeting, December 2026
21. IPS-25, the 25th International Conference on Photochemical Conversion and Storage of Solar Energy, Seoul, July 2026
20. 6th tandemPV Workshop, Berlin, June 2026
19. MRS Spring Meeting, April 2026
18. nanoGe – MATSUS Fall 2025, October 2025
17. Lawrence Symposium on Epitaxy, Arizona State University, October 2025
16. The University of Toledo, October 2025
15. MRS Spring Meeting, April 2025
14. Ludwig Maximilian University of Munich, March 2025
13. Nanyang Technological University, March 2025
12. National University of Singapore, February 2025
11. 4th tandemPV Workshop, June 2024
10. Physics Seminar Series, UC Merced, October 2023
9. 2nd Northwestern/Münster Symposium on Smart Materials, August 2023
8. Homeland Defense & Security Information Analysis Center, April 2023
7. APS March Meeting, March 2023
6. EcoMat Webinar: Perovskite Materials for Photovoltaic and Optoelectronic Applications (virtual), January 2023
5. Lawrence Symposium on Epitaxy, Arizona State University, January 2023
4. Zhejiang University, June 2022
3. KAUST Research Conference: 2022 Accelerating Solar Energy Research towards meeting Vision 2030 Goals, May 2022
2. ICFO – UofT – Stanford International School on the Frontiers of Light, October 2021
1. MRS Spring Meeting (virtual), April 2021

CONFERENCE ORGANIZATION & SERVICE

7. Organizer, nanoGe MATSUS Spring Meeting Symposium: Multilayer Perovskite Optoelectronics, Barcelona, Spring 2027
6. Organizer, ACS Fall Meeting Symposium: Materials Chemistry for Space Power Technologies, August 2026
5. Organizer, Northwestern–HZB–CU Boulder Tandem PV Workshop, May 2026
4. Organizer, ACS Spring Meeting Symposium: Organic, Perovskite, and Hybrid Solar Cells, March 2026

3. Organizer, MRS Spring Meeting Symposium, April 2025
2. Session chair, MRS Fall Meeting, November 2023
1. Organizer, ACS Fall Meeting Symposium: Organic, Perovskite and Hybrid Solar (raised $5,000 sponsorship for the symposium), August 2023

SELECTED ORAL & POSTER PRESENTATIONS

7. Oral, MRS Fall Meeting, December 2026
6. Oral, MRS Fall Meeting, December 2024
5. Oral, ACS Fall Meeting, August 2024
4. Oral, MRS Fall Meeting, November 2023
3. Oral, IEEE PVSC 50th, June 2023
2. Poster, MRS Spring Meeting, March 2017
1. Poster, MRS Spring Meeting, March 2016

TEACHING AND MENTORING

Faculty mentor, Kellogg-Q Entrepreneurial Residency 2024
Kellogg School of Management and the Querrey InQbation Lab, Northwestern University. Hosted MBA students embedded in our laboratory through this independent-study residency (Spring and Summer 2024), mentoring them one-on-one on the commercialization pathway for the perovskite photovoltaic technologies developed in the group; the residents presented their analyses to the full residency cohort and faculty.

Independent-study supervisor, ISP 398 Spring 2024
Weinberg College of Arts and Sciences, Northwestern University. Supervised an undergraduate independent study on solar cell physics and optoelectronic materials: designed the syllabus and reading plan, and guided the student from fundamentals to current research literature.

Independent-study supervisor, Mechanical Engineering 399 Fall 2024
McCormick School of Engineering, Northwestern University. Supervised an undergraduate independent project on the life-cycle assessment of triple-junction perovskite solar cells, guiding the student through the sustainability analysis of an emerging photovoltaic technology.

Postdoctoral and Ph.D. mentoring, Northwestern University 2022 – present
I direct the optoelectronics group, currently mentoring 16 postdoctoral fellows and research staff, and 8 Ph.D. students.
Current postdoctoral fellows and research staff: Linrui Duan, Hangyu Gu, Jin Hou, Xin Jiang, Saikiran Khamgaonkar, Chun-Hsiao Kuan, Chu Li, Shunde Li, Zheng Liang, Pengju Shi, Mengru Wang, Yurui Wang, Lewei Zeng, Xianfu Zhang, Yu Zhang, Qilin Zhou.
Current Ph.D. students: Isaiah Gilley, Chuying Huang, Kefu Huang, Ubaid Kazianga, Samantha Reitz, Matthias Schmidt, Taylor Wiggins, Jiale Zhuang.
Alumni placements: Dr. Somin Park (faculty, National University of Singapore); Dr. Cheng Liu (faculty, Shanghai Jiao Tong University); Dr. Yi Yang (faculty, Shanghai Jiao Tong University); Dr. Hao Chen (faculty, Shanghai Jiao Tong University); Dr. Chongwen Li (faculty, Southeast University); Dr. Guixiang Li (faculty, Southeast University); Dr. Yoon Jung Lee (faculty, Kookmin University); Dr. Deokjae Choi (Senior Researcher, Korea Institute of Science and Technology); Dr. Yuan Liu (Kadanoff-Rice Postdoctoral Fellow, University of Chicago); Dr. Donghoon Shin (Sejong Science Fellow, KAIST); Dr. Stefan Zeiske (promoted to Research Assistant Professor, Northwestern University).

Perovskite Photovoltaic Research Group Leader, University of Toronto 2019 – 2022
I manage a team of over 15 members focusing on perovskite photovoltaics research. My responsibilities

include mentoring on specific research projects, conceptualizing manuscripts, writing grant proposals, and preparing grant reports.

Lecturer (simulated), University of Toronto 2019

At the Teaching in Higher Education course at the University of Toronto, I developed my syllabus on "Two-dimensional semiconductor materials and systems" and taught in simulated classes.

ACADEMIC AND SOCIAL SERVICE

Journal advisory board 2025 – present

EES Solar

Journal reviewer 2016 – present

Science, Nature, Nature Energy, Nature Photonics, Nature Sustainability, Chemical Reviews, Journal of the American Chemical Society, Joule, Angewandte Chemie; Nature Communications, Advanced Materials, Matter, Energy & Environmental Science, Advanced Energy Materials, ACS Energy Letters, ACS Nano, ACS Photonics, Advanced Science, Chemical Science, EES Solar, Journal of Materials Chemistry C, ACS Applied Materials & Interfaces, ACS Applied Energy Materials; The Journal of Physical Chemistry, The Journal of Physical Chemistry Letters, Journal of Applied Physics, Progress in photovoltaics, Journal of Photovoltaics, Journal of Physics D: Applied Physics, Journal of Physics: Condensed Matter, 2D Materials, Nanotechnology, Nano Energy

Proposal reviewer 2023 – present

U.S. National Science Foundation
Natural Sciences and Engineering Research Council of Canada
ACS Petroleum Research Fund
National Research Foundation of Singapore
Ministry of Education, Singapore (Academic Research Fund Tier 3)

HONORS AND AWARDS

4. Stanford/Elsevier's list of top 2% scientists worldwide
3. Highly Cited Researcher in the field of Cross-Field - 2023 (Clarivate)
2. Highly Cited Researcher in the field of Cross-Field - 2024 (Clarivate)
1. Highly Cited Researcher in the field of Cross-Field - 2025 (Clarivate)

ALL PUBLICATIONS

144. Shin, D.†; Kitade, S.†; Duan, L.†; Liu, C.†; Li, C.†; Huang, C.; Kazianga, U. H.; López-Arteaga, R.; Reitz, S. A.; Choi, D.; Yang, Y.; Kim, H. J.; Lee, J.; Beights, J.; Kwon, H. W.; Song, J.-Y.; Liu, Y.; Choi, J.; Mirkin, C. A.; Kanatzidis, M. G.; **Chen, B.***; Sargent, E. H.* Sterically Gated Lewis Acid and Base Pairs Enable Orthogonal Defect Passivation in Perovskite Solar Cells. *Nat. Chem.* **2026**, in press. https://doi.org/10.1038/s41557-026-02261-z.

143. Shi, P.†; Zhuang, J.†; Zhang, D.†; Li, C.†; Zeiske, S.; Hou, J.; Gilley, I. W.; Yavuz, I.; Kazianga, U.; Li, C.; Zhang, Y.; Huang, K.; Itsoponpan, T.; Jiang, X.; Chen, C.-H.; Huang, C.; Yang, Y.; Zhang, X.; Nandi, P.; Reitz, S. A.; Facchetti, A.; White, K. P.; Wyatt, K.; Toney, M. F.; Kanatzidis, M. G.; Marks, T. J.*; Liu, C.*; **Chen, B.***; Sargent, E. H.* Phase-Homogeneous Mixed Halide Perovskites for Stable Tandem Photovoltaics. *Nature* **2026**. https://doi.org/10.1038/s41586-026-10929-2.

142. Ban, H. W.; Li, X.; Zeiske, S.; Steele, J. A.; López-Arteaga, R. E.; Lu, H.; Zou, Y.; Han, M. G.; Kim, T.-G.; Liu, C.; **Chen, B.**; Sargent, E. H. Short-Chain Acids Sustain InAs Colloidal Quantum Dot Growth during Synthesis, Extending Spectral Response into the Deep Short-Wave Infrared. *J. Am. Chem. Soc.* **2026**, *148* (21), 21885–21894.

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